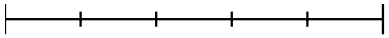
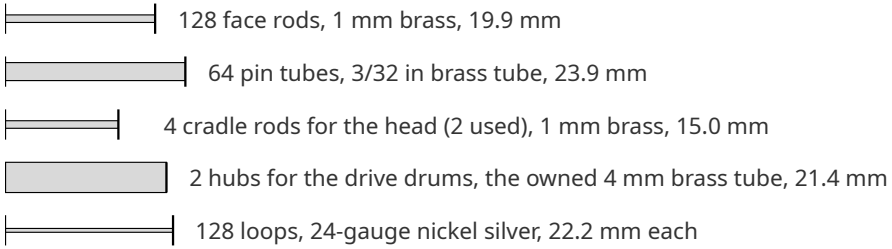
Tread and drum, actual size

Print at Actual size, then check the 50 mm bar with a ruler. Lay rods on the shapes to cut them, and build on the strip.



50 mm

1. Cut



128 face rods, 1 mm brass, 19.9 mm

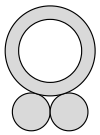
64 pin tubes, 3/32 in brass tube, 23.9 mm

4 cradle rods for the head (2 used), 1 mm brass, 15.0 mm

2 hubs for the drive drums, the owned 4 mm brass tube, 21.4 mm

128 loops, 24-gauge nickel silver, 22.2 mm each

2. Solder a plank: two rods side by side, the tube on top

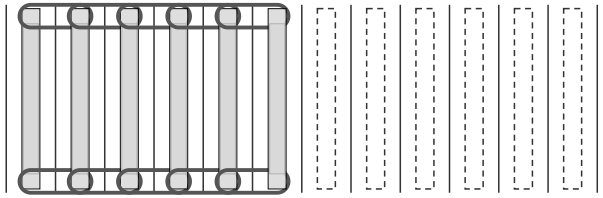


end on, 5 × size: 2.4 wide, 3.3 thick



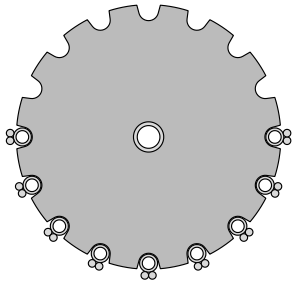
from the tread face, 2 × size: the tube stands out 2 mm each side for its two loops. Solder the rods to the tube along their length; keep the tube's ends bare.

3. Tape planks face down on the stations and close a loop round each pair of tube ends



6.5 mm stations: 32 planks close each track, the last loop closed on the rover. Each loop: 22 mm of 24-gauge nickel silver wrapped round the two pins, ends twisted, a dot of solder on the twist only.

4. Check a drum and a strip against this



Aluminium drum, two 3/8 in halves glued: 16 seats, 35.0 mm over the teeth. Each plank's pin tube sits in a round seat, 0.12 mm loose; the planks are 6.52 mm apart on the runs. Lay the strip round the drum: every pin tube should drop into a seat.

Bare aluminium, no paint: file each seat and the hole.

Drives on brass hubs; idlers turn on the 4 mm axle.

Rev K for approval, 2026-11-10. Frame: 3 mm tube, rails 64, rungs 65, back bar 15; 4 mm axle 113.5; four 5 mm collars 2. Numbers from build_page.py.